Implementation of counter using Timer

LAB # 08

Spring 2024

CSE-307L Microprocessor Based System Design Lab

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"On my honor, as student of University of Engineering and Technology, I have neither given nor received unauthorized assistance on this academic work."

Submitted to:

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Date:

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Task 1:

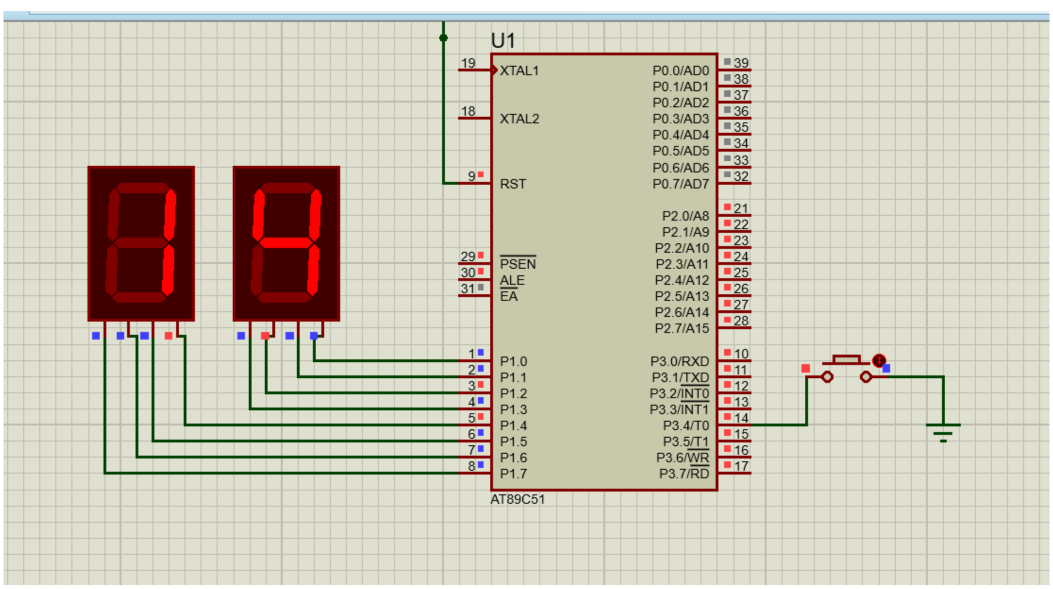
Implement Counter using Timer that take pulses from P3^4

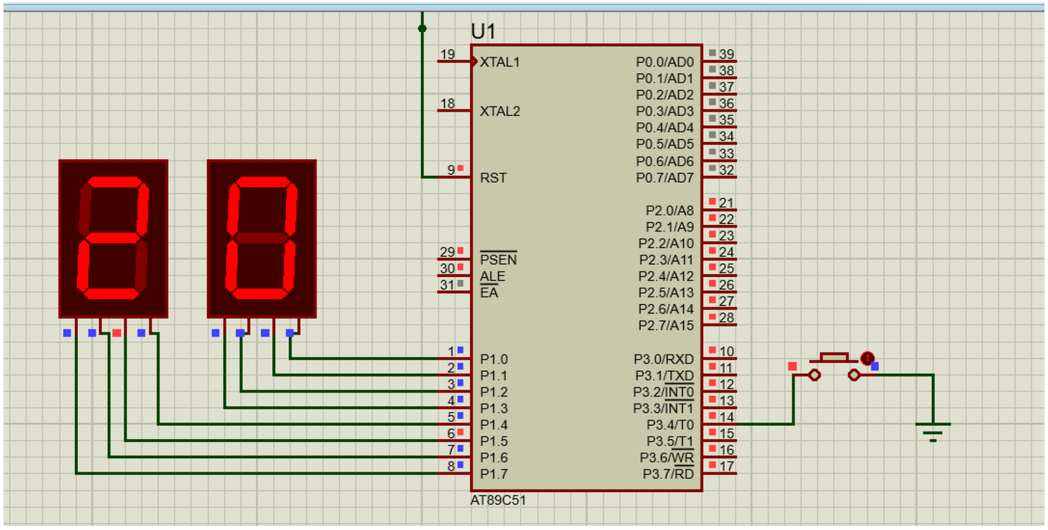
Code:

lab_task.c

```c
#include<reg51.h>

sbit btn = P3^4;
int unit, ten;

void main(void){

	//start_timer();
	TMOD = 0x06; // 0000 0110
	TH0 = 0x00;
	TL0 = 0x00;
	TR0 = 1;   //Part of TCON reg

	while(1){
	  unit = TL0 % 10;
	  ten= TL0 / 10;

	  P1 = (ten << 4) | unit;
	}
}
```

Proteus Simulation:





Task 2:

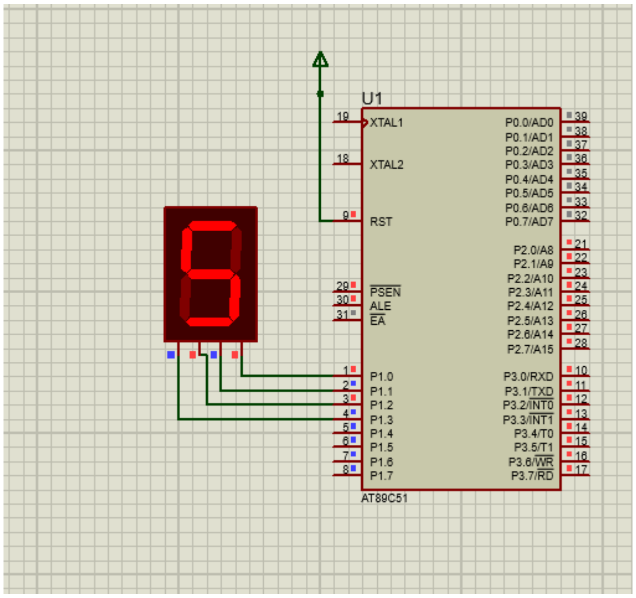
Implement Counter that counts from 0 to 9.

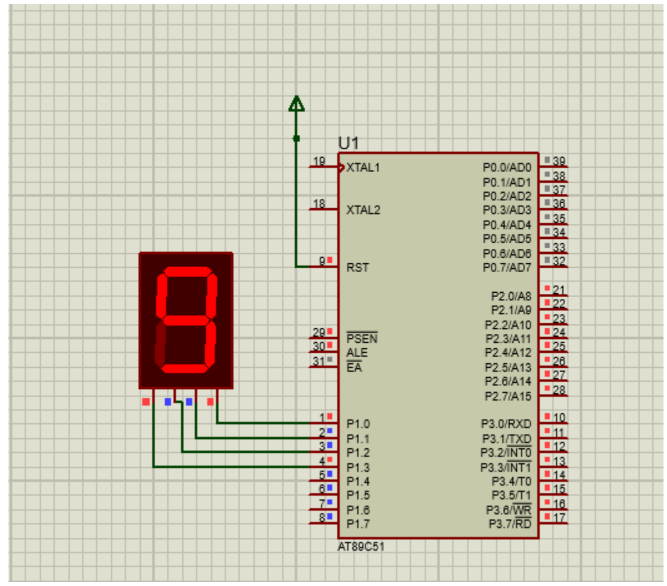
Code:

task2.c

```c
#include<reg51.h>

int i,k,l;

void delay(int limit){
        for(k = 0; k < limit; k++){
            for(l = 0; l < 1000; l++){}
        }
}

void main(void){
      while(1){
        for(i = 0; i < 10; i++){
              P1 = i;
              delay(100);
        }
      }
}
```

Proteus Simulation:





Task 3:

Implement Counter that counts from 00 to 99

Code:

task3.c

```c
#include<reg51.h>

sbit btn = P3^4;
int i,j,k,l;

void delay(int limit){
        for(k = 0; k < limit; k++){
            for(l = 0; l < 1000; l++){}
        }
}

void main(void){

      while(1){
        for(i = 0; i < 10; i++){
            for(j = 0; j < 10; j++){
            //     P1 = i*10 + j;
                // Convert to BCD
                P1 = (i << 4) | j;
                delay(100);
            }
        }
      }
}
```

Proteus Simulation: